# The Jacobian Conjecture after the Three-Dimensional Counterexample: Exact Validation, a General Seed Family, the Geometry of Non-Properness, and Two-Dimensional Equivariant Rigidity

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CAOS Research program, `github.com/fsantibanezleal/CAOS_RESEARCH`

11 November 2026

**Preprint** · v0.14, 11 November 2026 · Licence CC BY 4.0
CAOS Research open-problems programme · problem `jacobian-conjecture`
Version DOI 10.5281/zenodo.24226224
Concept DOI (always latest) 10.5281/zenodo.21503365
Not peer reviewed. Source code, data and computational records:
github.com/fsantibanezleal/CAOS_RESEARCH

**Abstract**

On 2026-07-20, Levent Alpöge announced (crediting the language model Claude Fable 5 and a question posed by Akhil) an explicit polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$ with constant Jacobian determinant $-2$ and a repeated value, refuting the Jacobian conjecture for every $N \geq 3$. We report an exact-arithmetic study of this map. First, we validate the counterexample independently and compute the full fiber over the announced target: it consists of exactly the three announced points. Second, we reverse-engineer the structure of $F$: it is a weighted skew-product over the invariants $v = xy$, $t = x^2 z$, its Keller property reduces to a two-variable identity, and its fibers are governed by a cubic in a single composite variable. Third, we derive and verify a general constructor: every admissible triple (seed polynomial $p$, scale $k$, section $q$) yields a Keller map with determinant $-k\,p(1)^2$ and generic fiber degree $\deg p + 1$; we exhibit new explicit counterexamples with rational collision certificates in fiber degrees 4, 5 and 6, plus a section-tail instance beyond the announced family shape, prove the degree law $(5d-3, 5d-4, 4)$ within the family, and show the announced map is its smallest member. Fourth, we make the failure geometry exact: a preimage escapes to infinity precisely when its fiber root is a multiple root, so the asymptotic variety of $F$ is the plane $C = 0$ together with the explicit discriminant surface $27A^2C^2 - 18ABC + 16A + B^3C - B^2 = 0$; over $\mathbb{R}$ this wall separates a three-sheet from a one-sheet region, and $F$ restricts to a surjective, non-injective real Keller map. The counterexamples reduce at suitable primes to non-injective Keller maps over $\mathbb{F}_\ell$ of degree below the characteristic. Fifth, we prove that the mechanism cannot reach the still-open two-variable case: for opposite-sign weights every $\mathbb{G}_m$-equivariant Keller map of $\mathbb{C}^2$ is linear (a classification with a positive-factor kill at the top degree), and with one zero weight it is affine, $(f_0 x,\, g_0 y + c)$, so a planar counterexample, if any, must be non-equivariant. Shaska's contemporaneous independent all-signature

classification is reconciled explicitly: same-sign actions may give nonlinear triangular automorphisms, while equivariant planar Keller maps remain automorphisms in every signature [10]. All computations are exact (over $\mathbb{Q}$, over finite fields, or symbolic) and reproducible from the repository; every claim below is labeled as machine-verified, derived, or conjectural.

# 1 Introduction

Let $k$ be a field of characteristic 0. A polynomial map $F = (F_1, \dots, F_N) : k^N \to k^N$ is a *Keller map* if $\det JF$ is a nonzero constant. The Jacobian conjecture $JC(N)$, stated in two variables by Kraus in 1884 (with a flawed proof whose gap, control of ramification at infinity, remained the subject's central obstacle [6]) and popularized by Keller in 1939 [4], asserted that every Keller map has a polynomial inverse. Over $\mathbb{C}$, invertibility is equivalent to injectivity, so a refutation requires exactly one repeated value. The classical positive ladder includes Wang's degree-2 theorem, Moh's analysis discarding planar counterexamples of degree up to 100 (with complete published detail only for the largest exceptional case; the current degree floor is discussed in the planar companion paper [7]), the Bass–Connell–Wright reduction to cubic homogeneous form [2], and Drużkowski's cubic-linear form [3]; Pinchuk [5] refuted the strong real variant by a non-proper local diffeomorphism with non-constant Jacobian. The conjecture is Problem 16 of Smale's list [9].

This paper studies the three-dimensional counterexample itself: validation, structure, the seed family, the geometry of non-properness, the real census, characteristic $p$, the weighted landscape, and two-dimensional equivariant rigidity as the bridge to the planar case. Two companion papers continue it. The planar paper [7] develops the $JC(2)$ machine built on the linearity of the Keller condition in one component, the uniform closure of the $\min\deg \le 2$ columns with one closed inverse, the unconditional Newton-polygon exclusion theorems and the annihilation lemma, the diagonal-corner reduction and the staircase transport, and a recalibration of the gcd and degree-floor landscape against the primary sources. The cascade paper [8] derives the consequences of the counterexample for related conjectures, each implication located in its primary source except Mathieu's 1997 paper, which was not located (that link rests on Zwart's review), and gives an explicit dimension-48 symmetric witness.

On 2026-07-20, Alpöge announced the map, with $u = 1 + xy$,

$$F = \big( u^3 z + y^2 u(4 + 3xy),\ \ y + 3xu^2 z + 3xy^2(4 + 3xy),\ \ 2x - 3x^2 y - x^3 z \big), \tag{1}$$

with $\det JF \equiv -2$ and $F(0, 0, -\frac{1}{4}) = F(1, -\frac{3}{2}, \frac{13}{2}) = F(-1, \frac{3}{2}, \frac{13}{2}) = (-\frac{1}{4}, 0, 0)$ [1]. Appending identity coordinates refutes $JC(N)$ for all $N \ge 3$; the case $N = 2$ remains open. The announcement consists of the map and its collision, both machine-checkable, and every property of $F$ used below is established by exact computation. Each computer-assisted result cites the archived computation that establishes it as EXP-*nnn* (see the section on reproducibility). Each claim is labeled: **[MV]** machine-verified exact identity, **[D]** derived analytically and certified on instances, **[C]** conjecture.

# 2 Exact validation and the full fiber (EXP-001)

**Theorem 2.1** ([MV])**.** *For $F$ as in (1): $\det JF = -2$ identically; the three announced points are pairwise distinct with the common image $(-\frac{1}{4}, 0, 0)$; and the fiber $F^{-1}(-\frac{1}{4}, 0, 0)$ consists of*

*exactly those three points. In particular $F$ is a non-injective Keller map and $JC(N)$ is false for every $N \geq 3$.*

The fiber computation is a lex Gröbner basis over $\mathbb{Q}$: the eliminant in $z$ is $\frac{1}{8}(2z-13)(4z+1)$, and on the fiber $3x+2y=0$, $12y^2 = 4z+1$. All coefficients of (1) are rational, so non-injectivity holds equally over $\mathbb{R}$.

## 3 The structure of the counterexample (EXP-002)

**Theorem 3.1** ([MV])**.** *$F$ is equivariant for the $\mathbb{G}_m$-action with source weights $(1,-1,-2)$ and target weights $(-2,-1,1)$. Writing $v = xy$, $t = x^2 z$, $u = 1+v$, the reduced data*

$$a_1 = x^2 P, \qquad b_1 = xQ, \qquad c_1 = R/x = 2 - 3v - t$$

*belong to $\mathbb{Q}[v,t]$. Each component of $F$ is affine in $z$, with coefficient vector $(u^3, 3xu^2, -x^3)$. Moreover, the generating identity $a_1 = \frac{ub_1}{2} - u^2 + \frac{u^3 c_1}{2}$ holds.*

**Lemma 3.2** (Skew-product determinant lemma, [MV] with generic symbolic coefficients)**.** *For* any *polynomials $a_1, b_1, c_1$ in $(v,t)$, the map $(a_1/x^2,\ b_1/x,\ xc_1)|_{v=xy,\, t=x^2 z}$ satisfies*

$$\det JF \cdot c_1^2 = -J_2, \qquad J_2 = \det \frac{\partial(V', T')}{\partial(v,t)}, \quad V' = b_1 c_1,\ T' = a_1 c_1^2.$$

For $F$, $J_2 = 2c_1^2$ **[MV]**: the Keller property is a *two-variable* identity; the third dimension only carries weights. Fibers are governed by a single cubic: with $w = uc_1/2$ and $\Phi(w) = w^2 - w^3$,

$$\Phi(w) = \tfrac{1}{4}\left(w\,QR - PR^2\right) \quad \text{identically } \textbf{[MV]}, \tag{2}$$

so preimages of $(A,B,C)$ correspond to roots of $4\Phi(w) - wBC + AC^2 = 0$; at the announced target this degenerates to $\Phi(w) = 0$ with roots $\{0,0,1\}$, matching the three preimages' $w$-values $(1,0,0)$ **[MV]**. Injectivity dies by *non-properness*: a Keller map is a local diffeomorphism, so sheets merge only at infinity; this is Pinchuk's loophole and the modern face of Kraus's gap.

## 4 The seed family: a general counterexample constructor (EXP-003/004)

The identification $c_1 = h(v) - t$ with $h = p/w$ is a degree-2 coincidence: a constructor built on it is *provably unsolvable* for $\deg p \geq 3$ (EXP-003, an exact unsolvability statement). Analyzing this obstruction leads to the correct *potential form.*

**Theorem 4.1** (Constructor, [D], certified on instances [MV])**.** *Let $p(w) = \sum_{i=1}^{d} p_i w^i$ satisfy*

$$p(0) = 0, \qquad \int_0^1 p = 0, \qquad p(1) \neq 0, \qquad p'(1) \neq 2p(1),$$

*let $k \neq 0$, $m = -k\,p(1)$, and let $q(v)$ be any polynomial with $q(0) = k$ and $q'(0) = k\,(p(1) - p'(1))/(p'(1) - 2p(1))$, with arbitrary higher-order tail. Set $\Phi(w) = \int_0^w p$, $s = c_1 = q(v) - t$, $w = us/k$, and define*

$$V' = k^2 p(w) + ms, \qquad T' = wV' - k^2 \Phi(w), \qquad b_1 = V'/s, \qquad a_1 = T'/s^2.$$

*Then $a_1, b_1$ are polynomials in $(v, t)$, and $F_{p,k,q} = (a_1/x^2,\ b_1/x,\ xc_1)$ is a polynomial Keller map of $\mathbb{C}^3$ with*

$$\det JF_{p,k,q} = -\frac{m^2}{k} = -k\,p(1)^2 \quad (\textit{constant, for every admissible } (p,k,q)),$$

*generic fiber degree $d+1$, and the exact reconstruction inverse: from a root $w$ of the fiber polynomial $k^2\Phi(w) - (w\,BC - AC^2)$,*

$$s = \frac{BC - k^2 p(w)}{m}, \quad u = \frac{kw}{s}, \quad x = \frac{C}{s}, \quad v = u - 1, \quad t = q(v) - s, \quad y = \frac{v}{x}, \quad z = \frac{t}{x^2}.$$

*The announced map (1) is the instance $p = 2w - 3w^2$, $k = 2$, $q$ affine; at $d = 2$ this shape is rigid up to the scale $k$ **[MV]**.*

The three admissibility conditions are exactly the low-weight polynomiality constraints at the origin; the potential form makes $J_2 = m^2 s^2/k$ automatic, so *the Keller property costs nothing*: this is the structural reason the conjecture is false in dimension $\geq 3$.

**Theorem 4.2** (New explicit counterexamples, [MV])**.** *The following maps are non-injective polynomial Keller maps of $\mathbb{C}^3$, each certified by an engineered rational collision:*

| *Seed $p$* | $k$ | det | *Component degrees* | *Generic fiber degree* |
|---|---|---|---|---|
| $w - 2w^3$ | 3 | $-3$ | $(12, 11, 4)$ | 4 |
| $8w - 12w^2 + 4w^3 - 5w^4$ | 14 | $-350$ | $(17, 16, 4)$ | 5 |
| $2w - 3w^2$, *section tail* $v^2$ | 2 | $-2$ | $(8, 7, 5)$ | 3 |

*For the first: $(-\frac{1}{15}, 18, 5130)$ and $(\frac{1}{24}, -18, -10368)$ both map exactly to $(-54, -54, 1)$. Collisions are engineered by prescribing two rational roots $w_1 \neq w_2$ and solving the linear system $w_i\tilde{V} - \tilde{T} = k^2\Phi(w_i)$ for the target invariants.*

The section-tail instance changes the component-degree pattern at fixed fiber degree and determinant: the family is strictly larger than the seed-only family circulating informally. Every negative rational number is realized as a Keller determinant within the family.

**Theorem 4.3** (Degree laws and family-internal minimality, [MV] for $d \leq 5$)**.** *For affine sections $q$ the component total degrees of $F_{p,k,q}$ are exactly $(5d-3,\ 5d-4,\ 4)$, $d = \deg p$ (verified $d = 2, \ldots, 5$; the $d = 5$ instance, seed $w - 3w^5$, $k = 5$, det $= -20$, degrees $(22, 21, 4)$, fiber degree 6, carries its own rational collision certificate). The scale $k$ never changes degrees; section tails strictly increase them; degree-1 seeds are impossible (the integral condition forces $p = 0$), so the fiber-degree floor is 3. Consequently the minimal degree vector over the family is $(7, 6, 4)$, attained exactly at the announced map:* the July 2026 counterexample is the smallest instance of its own family. *Globally, only Wang's degree-2 theorem constrains $\mathbb{C}^3$; the existence of a counterexample of total degree 3 to 6 remains open.*

*Remark* 4.4 (Characteristic $p$, [MV]). The $d = 3$ instance reduces mod $\ell = 13$ and $\ell = 101$ to explicit non-injective Keller maps over $\mathbb{F}_\ell$ of total degree $12 < \ell$ (the rational collision reduces to distinct points with a common image, e.g. mod 13: $(6, 5, 8)$ and $(6, 8, 6)$ both to $(11, 11, 1)$). Unlike the classical char-$p$ example $x - x^p$ (degree $= p$), the degree sits strictly below the characteristic, as expected from standard integer-model reductions; here it is certified explicitly.

# 5 The asymptotic variety: where sheets merge

**Theorem 5.1** (Escape = multiple fiber root, [MV] on $F$ and the $d = 3$ instance)**.** *In the potential form, the fiber polynomial $W(w) = k^2\Phi(w) - wBC + AC^2$ satisfies $W'(w) = k^2 p(w) - BC$ identically, and the reconstruction scale is $s = -W'(w)/m$. Hence a fiber root fails to reconstruct a finite preimage (the preimage escapes to infinity) precisely when it is a* multiple *root of $W$: over $C \neq 0$ the non-properness locus is the discriminant locus of $W$. For the announced $F$,*

$$A(F) \;=\; \{C = 0\} \;\cup\; \{\, 27A^2C^2 - 18ABC + 16A + B^3C - B^2 = 0 \,\}.$$

*Verified end to end: the target $(0, 1, 1)$ lies on the surface and its exact fiber is the single point $(2, -\frac{1}{2}, \frac{9}{8})$ (two sheets escaped); over generic $(A, B, 0)$ the fiber is the single flat-sheet point $(0, B, A - 4B^2)$; the collision target $(-16, -16, 1)$ lies off the surface with a full 3-point rational fiber. Collisions are generic; escapes are codimension 1. Kraus's "ramification at infinity" is, for this family, the discriminant of a univariate polynomial.*

# 6 The two-dimensional frontier: obstruction, rigidity, real census (EXP-005/006/010/011)

**Theorem 6.1** ([MV] with generic symbolic $f, g$)**.** *Let $\mathbb{G}_m$ act on $\mathbb{C}^2$ with weights $(1, -a)$, $a \geq 1$, and let $F = (x^{b_1} f(v),\, x^{b_2} g(v))$, $v = x^a y$, be an equivariant polynomial map. Then*

$$\det JF = x^{\,b_1+b_2+a-1}\left(b_1 f g' - b_2 f' g\right),$$

*so the Keller condition forces $b_1 + b_2 = 1 - a$ and the single univariate identity $b_1 f g' - b_2 f' g =$ const $\neq 0$.*

**Proposition 6.2** (One-invariant collapse)**.** *If the reduced data $(V', T')$ of a weighted skew-product depend on a single variable, then $dV' \wedge dT' = 0$, hence $J_2 \equiv 0$ and $\det JF \equiv 0$: no Keller map arises. Since a nontrivial $\mathbb{G}_m$-action on $\mathbb{C}^2$ has one independent weight-0 invariant (against two in $\mathbb{C}^3$), the potential-form mechanism of Theorem 4.1 is structurally unavailable in dimension 2.*

Additionally **[MV]**: no coordinate slice of (1) (each variable frozen at rational test values, any output pair) is a 2D Keller map.

**Theorem 6.3** (2D equivariant rigidity, [D]; for $w_2 \geq 1$ every mechanical step [MV] on sweeps)**.** *For any $\mathbb{G}_m$-action on $\mathbb{C}^2$ with weights $(w_1, -w_2)$, $w_1 \geq 1$, $w_2 \geq 1$, every equivariant polynomial Keller map is linear: $(f_0 x,\, g_0 y)$ or $(f_0 y,\, g_0 x)$. For weights $(w_1, 0)$, $w_1 \geq 1$, every equivariant polynomial Keller map is affine, $(f_0 x,\, g_0 y + c)$ with $f_0 g_0 \neq 0$, and it is linear exactly when $c = 0$.*

*Proof shape.* Weights $(w_1, 0)$ **[D]**: a monomial $x^i y^j$ has weight $iw_1$, so the component of weight $w_1$ is $P = xf(y)$ and the component of weight 0 is $Q = g(y)$. Then $\det JF = f(y)\, g'(y)$, which is a nonzero constant exactly when $f = f_0 \neq 0$ is constant and $g = g_0 y + c$ with $g_0 \neq 0$; the map $(x, y + 1)$ shows that the translation occurs. The rest of the proof concerns $w_2 \geq 1$. Equivariant components are $P = x^{i_1} y^{j_1} f(v)$, $Q = x^{i_2} y^{j_2} g(v)$ with $v = x^{w_2} y^{w_1}$, and the determinant factors as

$$\det JF = x^{i_1+i_2-1}\, y^{j_1+j_2-1}\big[(i_1 j_2 - i_2 j_1)\, fg + v\,(\beta_1 f g' - \beta_2 f' g)\big], \qquad \beta_r = w_1 i_r - w_2 j_r$$

(certified on 648 weight-times-exponent cases with generic symbolic $f, g$). Constancy forces $i_1 + i_2 = j_1 + j_2 = 1$, and the mixed shapes ($P = xyf$, $Q = g$ and its swap) admit no Keller solution (their bracket is divisible by $v$; empty solves). The surviving shapes reduce to $fg + v(w_1 fg' + w_2 f'g) = c$ (or its swap), whose coefficient at the top bidegree $N = d_f + d_g$ is $(1 + w_1 d_g + w_2 d_f)\, f_{d_f}\, g_{d_g}$ with a strictly positive factor: the leading coefficients die and by descent $f, g$ are constant (certified: empty constrained solves for all exact bidegrees $\leq 5$ over eight weight pairs; the pair $(1, 0)$ among them misses the translation $c$ through EXP-010's shape normalization, which is why the case $w_2 = 0$ is proved separately above). □

The independent enumeration EXP-006 (216 polynomial Keller instances over a bounded window, all 648 exactly-computed in-image fibers single-point, every instance linear **[MV]**) is the experimental companion of Theorem 6.3. Consequence: a two-variable counterexample, if one exists, must be non-equivariant: the symmetry class behind the entire three-dimensional family is completely closed in dimension 2.

**Scope and independent classification.** The word "linear" in Theorem 6.3 is restricted to the displayed opposite-sign scope; with one zero weight the maps are affine (the second statement of the theorem), in agreement with the classification below. Same-sign actions admit nonlinear triangular equivariant automorphisms. Shaska [10], in work submitted on 2026-07-22, independently proves the full classification: triangular for same-sign weights, affine when one weight is zero, and linear for opposite-sign weights. We make no priority or novelty claim for the classification. Our contribution here is the independently derived determinant factorization and the exact computations on the hyperbolic scope relevant to the three-dimensional mechanism.

**Theorem 6.4** (Real fiber census of $F$, [MV])**.** *Over* $\mathbb{R}$*, the announced* $F$ *restricts to a surjective, orientation-reversing (*$\det = -2$*), non-proper, non-injective real Keller map. For real targets off the discriminant wall the number of real preimages is 3 (region* $D > 0$*, which contains the collision targets) or 1 (region* $D < 0$*); two independent exact routes (Sturm isolation of the fiber cubic plus reconstruction, versus Gröbner elimination) agree at all samples, and no empty real fiber exists on a 36-target rational grid (an odd-degree argument makes this structural). In particular the constant-Jacobian* real *statement in dimension 3 falls with the same example; Pinchuk's planar counterexample (1994) concerned the non-constant case.*

The remaining routes to a hypothetical two-dimensional counterexample are: no symmetry at all; non-reductive ($\mathbb{G}_a$) structures, which face the same one-invariant collapse in naive form; or gradings at infinity in the Newton-polygon/Abhyankar sense. Conversely, a proof of $JC(2)$ in these terms is a properness statement: the reconstruction inverse of Theorem 4.1 shows exactly how asymptotic values are produced in dimension 3 (denominators $BC = k^2 p(w)$). Whether one-invariant geometry in the plane leaves room for such an escape is the question that the three-dimensional mechanism leaves open for $JC(2)$.

## 7 Discussion and open problems

**Why the conjecture is false.** In the equivariant class, being Keller is a two-variable potential identity, satisfiable for free at any fiber degree $\geq 3$; nothing in the determinant sees properness. **What is essential.** Two independent weight-0 invariants; the potential form; valuation conditions at the origin. **Consequence cascade (each implication located in its source, with**

**its dimension index checked, EXP-016, EXP-018 and EXP-136; Mathieu's 1997 paper was not located, and that link rests on Zwart's review).** The Mathieu conjecture is false for $SU(N)$, $N \geq 15$ (at fixed $N$, Mathieu's argument makes every Keller map $x - h$ of $\mathbb{C}^N$ with $h$ homogeneous invertible under the conjecture for $SU(N)$, and de Bondt's homogenization of the counterexample on $\mathbb{C}^{15}$, of degree 7 and verified exactly in EXP-138, is not injective) and for $SU(2)$ (an explicit example of Long, 2026); for $3 \leq N \leq 14$ it is not decided, since $F$ is not affinely of that form; abelian case Duistermaat–van der Kallen 1998. The Gaussian moments conjecture is false in every dimension $n \geq 3$ (GMC implies JC, Derksen–van den Essen–Zhao 2017; an explicit three-variable counterexample, Long 2026); Zhao's vanishing conjecture is false (equivalent to JC); the Image conjecture is false in some dimension (implies the vanishing conjecture); and the full Dixmier conjecture is false (Dixmier($n$) implies $JC(n)$, so Dixmier($n$) fails for every $n \geq 3$; the converse direction $JC(2n) \Rightarrow$ Dixmier($n$) of Belov-Kanel–Kontsevich and Tsuchimoto yields only stable equivalence, and ranks 1 and 2 remain open). The cascade is documented in the companion paper [8], together with an explicit dimension-48 symmetric witness (a gradient Keller map $\mathrm{id} + \nabla f$ with Hessian-nilpotent quartic potential and an exact two-point collision over $\mathbb{Q}(i)$, EXP-041). **Open problems.** In the plane, the companion paper [7] treats the transport closure of the fifth exclusion, the $B = 16$ normal form, and the $(72, 108)$ system. In dimension three, global minimality remains open: it is not known whether a Keller counterexample of total degree 3 to 6 exists (Theorem 4.3).

## The weighted landscape: uniqueness of the mechanism (EXP-012)

Generalizing to source weights $(1, -1, -m)$, the determinant lemma $\det JF = x^{\sigma+m} D$ and the pairing reductions $J_2 = \pm c_1^{r_a + r_b - 1} D$ hold generically **[MV]**, and the reduced Jacobian vanishes to order $m-1$ along the section. The survey outcome: at $m = 1$ the reduced pair is itself a planar Keller map, so that class is a *bridge* to $JC(2)$ (93 scanned instances, all injective; a collision there would refute $JC(2)$); at $m = 3$ the class $o = (-2, -2, 1)$ is *empty* (a one-line valuation proof, machine-certified) and $o = (-3, -1, 1)$ is rigid in scans (60 instances); at $m = 4$ the potential-form family does *not* exist: with seed degree up to 5 the admissibility system has Gröbner basis $[1]$ under $\mu \neq 0$ **[MV]**. In particular, a parity law producing families at every even $m$ does not hold: $m = 4$ admits no such family. More strongly, *within the scanned weighted landscape, the announced counterexample's weight system is the unique collision-capable mechanism.* Scope: sparse bounded scans, quantified in EXP-012; wider weight systems and higher-degree seeds are not covered.

## Reproducibility and data availability

Every theorem labeled [MV] is an exact sympy computation, over $\mathbb{Q}$ or, for the characteristic-$p$ statements (EXP-009), over $\mathbb{F}_\ell$, whose script and output are archived in the repository `https://github.com/fsantibanezleal/CAOS_RESEARCH`. An identifier EXP-*nnn* denotes one archived record in the experiment folders of the problem, `problems/algebraic-geometry/jacobian-conjecture/experiments/`; each record states its question, method and result, with the script and output of any computation. The results of this paper rest on EXP-001 through EXP-012, each of which also records an independent verification route; the Discussion also cites EXP-016 (the cited implications checked against their primary sources, except Mathieu's 1997 paper, which was not located) and EXP-041 (the dimension-48 witness).

## References

[1] L. Alpöge, announcement of the counterexample, X (@__alpoge__), 2026-07-20; question posed by Akhil, map produced with Claude Fable 5. Independent press coverage: M. Sparkes, *New Scientist*, 2026-07-20.

[2] H. Bass, E. H. Connell, D. Wright, The Jacobian conjecture: reduction of degree and formal expansion of the inverse, *Bull. Amer. Math. Soc. (N.S.)* **7** (1982), 287–330.

[3] L. M. Drużkowski, An effective approach to Keller's Jacobian conjecture, *Math. Ann.* **264** (1983), 303–313.

[4] O.-H. Keller, Ganze Cremona-Transformationen, *Monatsh. Math. Phys.* **47** (1939), 299–306. doi:10.1007/BF01695502.

[5] S. Pinchuk, A counterexample to the strong real Jacobian conjecture, *Math. Z.* **217** (1994), 1–4.

[6] L. O. Rodríguez Díaz, On the origin of the Jacobian conjecture, *C. R. Math.* **364** (2026), 363–370. arXiv:2512.23614.

[7] F. Santibáñez-Leal, Planar Keller maps after the counterexample: machine-certified Newton-polygon exclusions, the staircase transport, and the recalibrated frontier, preprint, 2026. doi:10.5281/zenodo.21503367.

[8] F. Santibáñez-Leal, The collateral impact of the three-dimensional Jacobian counterexample: the cascade of implications, preprint, 2026. doi:10.5281/zenodo.21503371.

[9] S. Smale, Mathematical problems for the next century, *Math. Intelligencer* **20** (1998), 7–15.

[10] T. Shaska, Graded Keller maps and the Jacobian Conjecture, arXiv:2607.20210v1, 2026.

[11] A. van den Essen, *Polynomial Automorphisms and the Jacobian Conjecture*, Progress in Mathematics 190, Birkhäuser, 2000.